Senior Comps Proposal:
Interactivity and Interpretability in Data Visualization Dashboards

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Abstract

Data visualization dashboards have become increasingly central to how people interpret information in various industries and sectors. Yet, the design and interaction choices embedded in these tools meaningfully shape what users conclude from the data. This project investigates whether interactive features, such as filtering, line isolation, sorting, and directional change indicators, improve users' ability to interpret time-series data compared with static visualizations that present the same information. Building on a four-week mini-project that compared static and interactive dashboards in Tableau Public using per-capita energy consumption data, this senior comprehensive project plans to address the methodological limitations of that initial study and broaden its scope. Specifically, the project will (1) recruit a larger and more demographically diverse participant pool, (2) perform a within-subjects methodology where each participant sees both versions, (3) develop a richer scoring rubric that credits depth of reasoning on open-ended tasks rather than only accuracy and speed, and (4) rebuild the dashboards in an open-source Python framework to move beyond Tableau-specific findings. The dataset will also shift from energy consumption to a higher-stakes domain, where misinterpretation carries real-world consequences. The aim is to produce findings about interaction and interpretability that are more credible, more generalizable, and more directly relevant to the kinds of decisions dashboards are increasingly used to support.

1 Introduction

Dashboards have moved from specialized analyst tools into everyday decision-making infrastructure. Doctors use them to triage patients [23, 35], HR employees use them to support workforce and staffing decisions [21], and journalists create and use them to communicate election results [3, 1]. The Johns Hopkins COVID-19 dashboard alone shaped how the public understood the pandemic, and later research showed that reasoning was sensitive to framing choices: log vs. linear scales, daily vs. cumulative counts, and how the map colors were grouped [25, 30]. They all used the same data but made different design choices, leading to different conclusions.

This raises a serious research question: when users are given interactive tools to explore data rather than a static chart of the same information, do they actually interpret it more accurately, or do they only feel like they do? The distinction matters. If interactivity primarily reduces the perceived effort of analysis without improving the conclusions users reach, then the widespread shift toward interactive dashboards may improve user experience while leaving interpretation quality unchanged, or even worse, giving users a false sense of confidence in their conclusions.

The mini-project found mixed results on this question. Across six participants comparing static and interactive Tableau dashboards of per-capita energy consumption [33], interactive users reported substantially lower cognitive load and higher ease of use. However, their accuracy was comparable to that of static users, and their average completion times were longer [29]. Additionally, half of the participants independently asked for a feature neither version provided: explicit indicators of year-over-year change [29].

The senior comprehensive project will build on these findings while addressing the mini-project's methodological limitations. The research questions guiding this work are:

- **RQ1**: Does interactivity with time-series data improve interpretation accuracy and reasoning depth compared to static visualization, or does it primarily reduce perceived cognitive load?
- **RQ2**: Do explicit directional change indicators, when added to either static or interactive dashboards, change which patterns users notice and what conclusions they draw?
- **RQ3**: Are findings about interactivity tied to a specific platform (Tableau), or do they generalize across implementations built in different frameworks?

2 Technical Background

Many widely used modern data visualization dashboards include interactive controls that allow users to filter, sort, drill down into, or highlight subsets of the data typically presented through charts, tables, and summary statistics. Tools such as Tableau, Power BI, and Looker dominate the commercial space, while open-source frameworks such as Plotly, Streamlit, and Dash are commonly used in research, software, and engineering contexts.

Interpretability in regard to data visualization refers to whether users can correctly extract the intended meaning from a visualization and reason about the underlying data. This is distinct from usability, which measures how easily a user can navigate an interface. A dashboard can be highly usable (easy to click around) while producing poor interpretability (users walk away with wrong conclusions), and the relationship between these two properties is one I try to test for this project.

Interaction techniques in visualization span greatly, and they can include but are not limited to: filtering (restricting the data shown to a subset), brushing and linking (selecting items in one view to highlight them across others), drill-down (moving from aggregate to detail views), sorting, and zooming [36].

Cognitive load refers to the amount of information our working memory can process at any given time [34]. The visualization community typically distinguishes between intrinsic load (the inherent complexity of the data and task), extraneous load (load imposed by poor design), and germane load (load contributing to learning and reasoning) [6]. Interactive features can, in principle, reduce extraneous load by letting users hide irrelevant information, but they can also add load by introducing controls that have a barrier to learning [17].

Time-series data consists of measurements of one or more observations across time. For my project, tasks revolved around this type of data will include value retrieval, identifying trends, comparing values across categories, detecting outliers, or making predictions based on observed patterns, as these tasks are grounded in established visualization evaluation practices [36, 2, 22].

For senior comps, a within-subjects design would involve each participant using the interactive and static visualizations, as opposed to the methodology I went with for my mini-project, a between-subjects design where each participant only used the interactive or static visualizations. Within-subjects designs are statistically more powerful for small samples because each participant serves as their own control; however, they also introduce risks of bias regarding order and learning effects that must be controlled for [12, 8].

3 Prior Work

The literature this project draws on falls into five connected areas: the role of interaction in reasoning, sensemaking and cognitive load, user variation in interpretation, accessibility, and methodologies for evaluating interpretability.

3.1 Interaction Driving Reasoning

An important distinction to be made is between interaction-for-usability and interaction-for-reasoning. Yi et al. define and categorize interaction techniques by user intent and argue that interaction lets users actively explore data rather than passively consuming it [36]. Dimara and Perin extended this by arguing that meaningful interaction should encourage users to question, compare, and engage with data rather than simply making interfaces faster to navigate [10]. Together, these papers establish that interactivity should be judged by whether it changes how users reason, not just how quickly they finish tasks, motivating my project's central question: if interactivity is supposed to support reasoning, then evaluations should measure reasoning depth, not just task accuracy/speed.

3.2 Sensemaking, Cognitive Load, and Visual Complexity

Similar literature has examined the cognitive processes users engage in while interpreting data. Pirolli and Card describe sensemaking as a process of turning information into meaning through an iterative cycle of foraging (searching for relevant information) and synthesis (building coherent insights) [26], while Klein et al. describe how users approach data with mental models/frames they continuously update as new information arrives [16]. Filtering and drill-down map onto these cognitive processes by allowing users to iteratively test and refine their understanding of the data: filtering supports foraging by allowing users to focus on a relevant subset of the data, and drill-down supports navigating between summary and detail.

But interaction itself imposes cognitive demands. Ware's foundational work on visualization perception argues that designers must actively avoid cognitive overload, which occurs when an individual is unable to process the amount of information being received [34], and Hullman shows that even well-intentioned additions like uncertainty indicators can overwhelm users, increase cognitive load, and hinder interpretation [13]. Empirical evidence shows that certain visual choices directly affect both accuracy and effort [11, 37]. The takeaway is that adding features, including interactive ones, is not a neutral representation of data; every addition trades off potential reasoning support against added cognitive demand [17]. The mini-project's results support the finding that interactive users took longer, despite reporting lower mental effort.

3.3 User Variation and Subjective Interpretation

Even when participants see the same visualization with the same data, they may all reach different conclusions. Quadri et al. found that users will often diverge from the intended interpretation and construct their own conclusions through their own background knowledge and experiences [27]. Choe et al. similarly found that users construct meaning based on prior beliefs, sometimes filtering visualized information to fit their pre-existing narratives and disregarding the visualized information that conflicts with their viewpoint [9]. These papers are important because they argue against scoring rubrics that treat interpretation as a single correct answer. A participant who reads energy data through the lens of colonial history is doing something more sophisticated than a participant who only spots the tallest bar, and rubrics that score only the tallest-bar answer miss this distinction. This directly motivates the development of a richer scoring rubric for this project.

3.4 Accessibility and the Limits of Default Designs

Srinivasan et al.'s work on screen-reader accessibility for dashboards argues that dashboards that prioritize the sighted lock blind and low-vision users out of data analysis; accessibility isn't an add-on but must be built in from the start [32]. The W3C Web Content Accessibility Guidelines outline standards for color contrast, keyboard navigation, screen reader support, and text alternatives for visuals that any responsible dashboard project should meet [14]. While full screen-reader accessibility is probably beyond the scope of this project, these sources inform a baseline of accessibility commitments that the project will adopt: choosing color palettes that meet WCAG contrast standards, ensuring interactive elements are keyboard-navigable, and acknowledging potential accessibility shortcomings.

3.5 Evaluating Interpretability

To objectively measure how data visualizations effectively support user understanding, decision making, and interpretability, researchers must empirically evaluate users through predefined analytical tasks [18]. Amar, Eagan, and Stasko provide low-level analytical tasks (retrieve values, find extrema, compare trends across categories, etc.) that are among the most common and well-validated task types in visualization evaluation that I plan to model after for the project [2].

Lam et al. argue for combining objective metrics with subjective measurements, such as the Likert-scale instruments, which provide validated approaches to capturing perceived workload and clarity [18]. The mini-project used a similar Likert scale; the senior project will continue this approach while adding more structured open-ended prompts informed by the insight-based evaluation literature [31]. North and Chang et al. push further, arguing that visualization effectiveness should be measured through the depth and quality of insights generated, not just accuracy [24, 7].

Taken together, interpretability is multifaceted, interaction shapes reasoning, users vary meaningfully in how they make sense of the same data, and good evaluation requires both objective and subjective measures combined with task designs that span multiple analytical levels [18]. The senior project is an attempt at that through a study that the mini-project, due to scope and time constraints, could not be.

4 Methods

4.1 Approach

The project will compare two dashboard versions presenting the same time-series dataset: (1) a static version with no interactive controls and (2) an interactive version that notes explicit year-over-year directional changes with filtering, sorting, and line isolation. The directional indicator has been added in response to mini-project participants independently requesting this feature.

Both versions will be built using Plotly Dash, an open-source Python tool, in addition to Tableau. This is a deliberate expansion from the mini-project. The dashboard tooling world is concentrated in a handful of big U.S. tech companies whose design decisions shape how the rest of the world interprets data. Having chosen Tableau Public for my mini-project, I was part of that concentration: my findings about what made dashboards interpretable were really findings about Tableau's version of interactivity, not interactivity in general [28]. By building visualizations with Plotly Dash, my project's design choices become independent from commercial enterprises, allowing me to showcase the underlying code to make the dashboards fully reproducible by anyone with Python for inspection and potential extensions. Visual design choices (chart type, color palette, axis scales, layout) will be held as constant as possible across the versions and tools so that any differences can be attributed to interactivity rather than to design.

The dataset choice will differ from per-capita energy consumption [33] to a more meaningful, higher-stakes sector. Why? Because it tests whether the patterns observed in the mini-project (interactivity reducing perceived cognitive load without improving accuracy) hold in a domain where misinterpretation has more serious consequences.

The study will revolve around a within-subjects design, where each participant will use the static and interactive versions in a randomized order, with comparable but distinct tasks to avoid memorization between versions. This methodology was an improvement I found from the mini-project's reflection, since between-subjects with six participants gave each version almost no statistical power [8]. With a target of 25+ participants interacting with both versions, each participant serves as their own control, increasing statistical power and reducing standard error relative to a between-subjects design with the same total sample.

4.2 Justification and Alternatives Considered

A controlled comparison between the static and interactive dashboards using the same data and design choices remains the most appropriate method for isolating the effect of interactivity, as prior research argues that it cannot be meaningfully assessed without controlling for other design variables [18, 10]. Task-based user studies are well established in visualization research for evaluating how users understand and reason [18], and the mini-project served as a proof of concept that it was possible to perform such a study.

Three alternative extensions to the project were considered and set aside for now. First, think-aloud protocols, where participants are told to verbalize their reasoning while interacting with the system, would provide some rich insight into the cognitive processes as they occur [5], but would be impractical to implement and scale to 25+ participants within the academic semester. Instead, an improved version of post-task open-ended written responses from the mini project will serve as a more feasible substitute that still captures qualitative reasoning. Secondly, eye-tracking would offer direct evidence of which parts of a visualization users actually look at, but the collection of biometric data raises consent, privacy, and data-handling concerns that may fall outside the scope of the project. Lastly, longitudinal insight-based evaluation, as proposed by Saraiya et al., would track how user understanding develops over weeks of dashboard use [31], but is not possible given the time constraints of the senior comps.

4.3 Materials

To complete this project, several resources will be required. The static and interactive dashboards will be built using two different platforms: a commercial vs. an open source platform. Tableau Public is a free visualization platform and the tool I used for the mini-project, as it lets me hold the visual design constant while changing only the interactivity. On the other hand, Plotly Dash (Python) is a free and open-source software commonly used for highly customizable and interactive web-based data visualizations and production-grade applications.

All code will be published in a public repository hosted on my GitHub profile to support accessibility and replication. Both dashboard versions (Tableau Public and Plotly Dash) will be embedded in a custom web application that handles the visualizations, task presentation, response collection, and automatic logging of task completion times and interaction events (clicks, filters, and selections). This addresses one of the mini-project's limitations: reliance on self-reported timing, which probably had timing errors in the seconds.

5 Evaluation Metrics

The evaluation uses four metric categories, refined from the mini-project based on what worked and what did not.

Task accuracy will be measured for clear-cut interpretation tasks (value retrieval, trend judgment, rank comparison) drawn from Amar, Eagan, and Stasko's work on low-level analytical activities [2]. These provide an objective measure of how effectively the system supports correct user understanding and decision making.

Reasoning depth is the most substantial addition from the mini-project. Open-ended trend questions will be scored using a multifaceted and structured rubric that credits: accuracy, evidence cited from the visualization, depth of comparison across categories, and engagement with possible alternative explanations. Quadri et al. motivate this extension as they argue that high-level interpretation is genuinely interpretive and that reductive scoring penalizes the kind of reasoning visualizations are supposed to support [27].

Task completion time will be logged automatically by the dashboard rather than self-reported, removing a known source of error in the mini-project data [29]. Completion time is included not as a primary outcome but as context for interpreting accuracy and reasoning results; a participant who takes their time but scores well may be using interaction productively, while one who blasts through but scores poorly may be missing patterns.

Subjective ratings will continue to use Likert-scale questions to measure perceived clarity, confidence, ease of use, and cognitive load, providing some qualitative insights into a participant's reasoning processes and potential sources of misinterpretation [15, 18]. The mini-project used these successfully and actually found the largest interactive vs. static differences here, supporting their continued use for comps.

Several alternatives were considered and set aside for now. Insight-based metrics were considered for the open-ended portion [24], but the structured rubric was preferred because it produces more reproducible scores. Eye-tracking-based attention metrics would give direct evidence of what users actually look at, but as noted earlier, the collection of biometric data raises consent, privacy, and data-handling concerns that may fall outside the scope of the project. Lastly, as mentioned earlier, longitudinal insight-based evaluation would track how user understanding develops over weeks of dashboard use [31], but is not possible given the time constraints of the senior comps.

6 Mini-Project Results and How the Senior Project Builds on Them

The four-week mini-project compared a static and an interactive Tableau Public dashboard of per-capita energy consumption, with six participants completing six interpretation tasks across the two versions (n = 2 static, n = 4 interactive) [29].

The clearest finding was a substantial reduction in self-reported cognitive load in the interactive condition. Ratings of ”the visualization felt overwhelming” and ”it was difficult to track multiple countries” dropped by 2.75 and 2.50 points, respectively, on a 7-point scale, and ease-of-use rose by 1.50 points. These are large effects despite the small sample and are aligned with what participants shared in the open-ended responses: static users described struggling to track individual countries through overlapping line segments, while interactive users described filtering and isolation as the techniques that made their reasoning manageable.

Two findings complicated the simple ”interactivity helps interpretability” statement. First, the mean task completion time was longer in the interactive condition (8 minutes 40 seconds versus 6 minutes 37 seconds), opposite the expectation that filtering would speed up comparison. The most likely explanations are either that interactive features invite and promote user exploration that lengthens engagement, or that interaction overhead, which is the time spent toggling, filtering, and re-filtering, exceeds the time saved on comparison itself. Second, accuracy was comparable between the static and interactive versions on straightforward, clear-cut tasks, with both groups answering value retrieval and trend judgment questions correctly nearly all the time, and both struggling with rank comparison (50% accuracy in both conditions). Self-reported confidence was marginally higher in the static condition (6.50 versus 5.75), possibly because static users had fewer ways to second-guess their initial reading.

The most actionable finding was qualitative: half of the participants independently asked for explicit numeric or directional indicators of year-over-year change, a feature neither version provided.

The senior comprehensive project extends the mini-project in four concrete ways, each addressing a specific limitation:

1. **Sample Size and Diversity**: Six undergraduates, predominantly computer science, are a known weakness; 25+ participants recruited more broadly address both the statistical power [19] and the demographic narrowness [20], which is consistent with typical sample sizes in HCI user studies[4].
2. **Within-Subjects Design**: Between-subjects with six participants meant essentially no statistical leverage; within-subjects increases statistical power and reduces standard error relative to a between-subjects design with the same total sample [8, 12].
3. **Reasoning Rubric**: The mini-project really only scored accuracy and speed; the senior rubric will also credit reasoning depth on open-ended responses [27].
4. **Open-Source and Higher-Stakes Domain**: Plotly Dash will be compared against Tableau findings, and a separate, more significant dataset will replace energy consumption, addressing both platform-dependence and stakes concerns [25]

The mini-project's finding of comparable accuracy with reduced cognitive load is interesting precisely because it is mixed; the senior project is designed to determine whether this pattern holds with better methodology, in a different platform, on a higher-stakes domain.

7 Ethical Considerations

The ethical issues this project raises were examined in greater depth in the ethics considerations paper preceding this paper [28]; this section summarizes those concerns and specifies what the senior project commits to addressing.

Sampling bias and whose interpretations count. The mini-project's sample of six participants aged 17–21 from a single college campus, predominantly computer science majors [29], cannot speak to how dashboards work for the populations actually using them: older adults, people without college degrees, non-native English speakers, and users in different national contexts [4]. The senior project commits to a larger and more demographically diverse sample (25+ participants recruited on campus and through external means) while being upfront that even this expanded sample has its limits. A subtler bias concern is whose interpretations get marked correct: scoring a participant who reads colonial-history patterns into energy data the same as one who only spots the tallest bar treats interpretation reductively [27]. The new reasoning rubric directly addresses this by crediting depth of reasoning, not just accuracy.

Accessibility. Tableau Public's accessibility is partial at best, and the mini-project did not include blind or low-vision participants in design or evaluation. The senior project cannot fully solve this. Meaningful inclusion of blind and low-vision users in design from the start, as Srinivasan et al. argue [32], is necessary but beyond the scope of my senior project. What I can commit to is a baseline of color palettes that meet WCAG contrast standards, ensuring interactive elements are keyboard-navigable, and acknowledging potential accessibility shortcomings [14] in the Plotly Dash implementation. As a result, I will have to explicitly state that the dashboards are designed for sighted users without major impairments. Building in Plotly Dash allows me to publish the underlying code in a public repository hosted on my GitHub profile, making the project accessible, inspectable, and modifiable, which a closed Tableau implementation doesn't natively support.

Platform concentration and ownership. The mini-project used Tableau Public, which is owned by Salesforce; Power BI is owned by Microsoft, and Looker by Google. Findings from any of these tools are findings about that company's specific implementation of interactivity, not about interactivity in general. Moving to Plotly Dash addresses this directly: the open-source framework removes the project from any single commercial company's design decisions, which largely try to appeal to the majority while overlooking the needs of users whose workflows and accessibility requirements fall outside the assumed mainstream.

Data domain and global inequity. The mini-project's per-capita energy dataset, drawn from *Our World in Data* [33], reflects the inequalities of where the data comes from. Countries with strong national statistics agencies (the U.S., U.K., and E.U. members) have richer and more accurate data than countries with weaker data infrastructure, and any dashboard built on uneven data inherits that unevenness. The senior project will document this clearly if this issue also applies to a new dataset, rather than treating the underlying data as neutral.

Higher stakes and researcher responsibility. Moving away from energy data raises the stakes of the project. If interactive dashboards genuinely help users interpret data more accurately, that is a meaningful contribution; if they primarily reduce perceived effort while leaving conclusions unchanged, that is a finding worth reporting, as it complicates the previously mentioned "interactivity helps interpretability" assumption. Either result is meaningful, so this project will report null and incorrect results with the same care as correct ones.

8 Schedule and Timeline

Summer 2026 (preparation). Finalize my dataset selection and obtain any necessary permissions or IRB approval. Prototype a working Plotly Dash implementation to make sure the tool is appropriate and sufficient for the project.

Weeks 1–3. Build both dashboard versions in Plotly Dash and Tableau Public so they look the same and use the same visual design choices, with the only difference being interactivity. Write the task questions and the scoring rubric, then have another person pilot-test the rubric on a small set of practice responses to make sure the directions and instructions are clear before the real study begins.

Weeks 4–6. Run a small-scale/early version of the full study with 3 to 5 participants to test timing, instructions, and logging. Make adjustments based on what the study reveals or feedback given. Once the IRB approves the study, start recruiting participants for the real version.

Weeks 7–10. Run the main study with 25+ participants, scheduling sessions throughout the period rather than all at once, so I can check the data early and catch any issues as the sessions progress.

Weeks 11–13. Run the quantitative analysis on accuracy, completion time, and Likert ratings; score the open-ended responses.

Weeks 14–15. Write the final paper, finalize replication instructions and code architecture appendix, and prepare for the comps presentation.

9 Conclusion

The mini-project suggested that interactivity reduced perceived cognitive load without clearly improving accuracy. Participants also wanted to see a feature that neither version offered be implemented: directional change indicators. The senior project responds to each of these issues directly. A within-subjects design with at least 25 participants gives the study enough statistical strength to find real differences. A scoring rubric credits the depth of participants' reasoning, not just whether they got the answer right or how fast they got there. The interactive version includes the directional indicators people asked for. And building the dashboard in both Tableau Public and Plotly Dash means I can actually test whether findings are platform-dependent, rather than just assuming they don't.

The choice to use a dataset from a higher-stakes industry, rather than the energy data from the mini-project, reflects why this question matters in the first place. Dashboards aren't just tools for data analysts anymore; they're how decisions get made in various sectors like healthcare, public policy, and journalism. Therefore, the design choices baked into these visualization tools shape what people walk away believing about the data.

If this project works as planned, it should produce evidence about interactivity and interpretability that is more credible than the mini-projects', more generalizable beyond a specific commercial tool, and more relevant to the kinds of decisions dashboards are increasingly used to support. If the mini-project's patterns hold up under a stronger study design, that is worth taking seriously. If they don't, that is equally worth knowing.

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